

Agentic AI: Building Autonomous Systems for Real-World Applications

Module 1: Foundations of Agentic AI & LLM Capabilities

Quick Reference Guide

Learning Outcomes

- Identify how large language models are constructed and the role of scaling laws in shaping model performance.
- Describe limitations of large language models, including fragility, scaling constraints, and inconsistent behavior across models.
- Define the core components of an agent, including goals, perception, memory, reasoning, and action.
- Distinguish between prompt-based model outputs and agent workflows that use structured reasoning steps.
- Define an initial agent concept for a real-world problem, including the task goal, environment, required reasoning steps, and potential agent structure.

Foundations of LLMs

- LLMs learn a probability distribution over token sequences and generate text through autoregressive next-token prediction.
- During training, models minimize cross-entropy loss to increase the likelihood of correct tokens given prior context.
- Transformer architectures replace recurrent computation with self-attention, enabling parallel processing of sequences.
- Self-attention computes pairwise relevance between tokens using learned query, key, and value representations.
- Positional encodings inject order information that transformer models do not inherently capture.
- Decoder-only architectures generate text one token at a time during inference using causal masking.

Scaling Laws and Model Limitations

- Scaling laws show that model loss decreases predictably as a power law function of data size, parameter count, and compute.
- Larger models trained on more data exhibit emergent capabilities such as improved reasoning and generalization.
- Attention mechanisms scale quadratically with context length, creating memory and compute constraints.

- Longer context windows expand the scope of reasoning but increase inference cost and system instability.
- LLMs remain fragile, meaning small changes in prompts can lead to significantly different outputs.
- Model behavior can vary across providers due to differences in architecture, training data, and context limits.
- Language models optimize likelihood rather than correctness, grounding, or real-world task success.

Agents and Agent Workflows

- A prompt response is a single forward pass through a language model with no memory, verification, or iteration.
- An agent embeds a language model within a system that operates over time.
- The model component generates candidate actions using probabilistic reasoning.
- The environment defines what the agent can observe and what actions it can take, including tools, data sources, and users.
- Actions allow the agent to affect the environment through tool calls, code execution, or data retrieval.
- Feedback provides signals about action outcomes and guides subsequent decisions.
- Agent workflows follow an observe, reason, act, and feedback loop rather than a single generation step.
- This structure enables error correction, iteration, and goal-directed behavior beyond text generation.
- Agent design addresses the limits of scaling by adding structure, tools, and feedback rather than relying on larger models alone.